



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: V.Madhulika

Reg. No.:192312620

Course Code: CSA1709

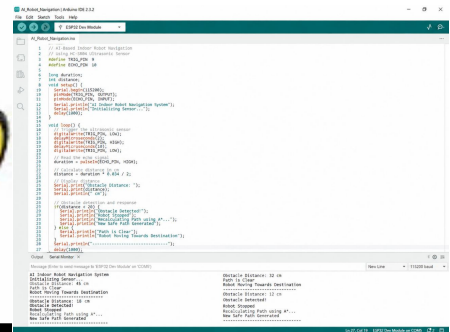
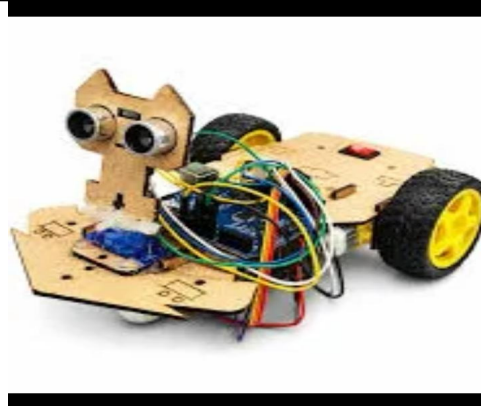
Slot: B

Course Name: Artificial Intelligence

Course Faculty: Dr.P.Rajaram

Project Title: AI Based Indoor Robot Navigation Using A* Search And Dynamic Obstacle Avoidance

Module Photographs:



Project Description: Module-1: Environment Mapping

The project “AI-Based Indoor Robot Navigation Using A Search and Dynamic Obstacle Avoidance”^{*} focuses on developing an intelligent robot navigation system that allows a robot to move autonomously in indoor environments such as classrooms, laboratories, offices, hospitals, warehouses, and smart homes. For safe and efficient navigation, the robot must first understand its surrounding environment and identify the locations of obstacles. In Module 1 – Environment Mapping, the robot uses the HC-SR04 ultrasonic sensor to scan and understand its surroundings. The sensor continuously measures the distance between the robot and nearby objects, helping the system detect obstacles present in the indoor environment.

The generated environment map serves as the input for the Path Planning Module. By providing information about the robot's surroundings and obstacle locations, Module 1 creates the foundation required for the A* Search Algorithm to determine a safe and efficient route from the starting point to the destination. Thus, Environment Mapping is the first stage of the proposed system, enabling the robot to understand its surroundings before beginning the navigation process.

V.Madhulika
Student Signature

Guide Signature